Expansiveness and the Anosov Closing Lemma

Lemma 1 (Expansiveness)**.** *Let*

$$f : M \to M$$

be a uniformly hyperbolic diffeomorphism. Then there exists

$$\varepsilon_0 > 0$$

such that

$$d(f^n(x), f^n(y)) < \varepsilon_0 \qquad \forall n \in \mathbb{Z}$$

implies

$$x = y.$$

Proof. Locally decompose

$$y - x = v^s + v^u, \qquad v^{s,u} \in E_x^{s,u}.$$

If

$$v^u \neq 0,$$

uniform hyperbolicity gives exponential expansion:

$$\|Df^n(x)v^u\| \geq C^{-1}\lambda^{-n}\|v^u\|, \qquad 0 < \lambda < 1,$$

so

$$d(f^n(x), f^n(y)) \to \infty$$

as

$$n \to \infty,$$

contradicting boundedness.
Similarly, if

$$v^s \neq 0,$$

then backward iterates satisfy

$$d(f^{-n}(x), f^{-n}(y)) \to \infty.$$

Hence

$$v^s = v^u = 0,$$

so

$$x = y.$$

□

Theorem 1 (Anosov Closing Lemma)**.** *Let*

$$f : M \to M$$

be uniformly hyperbolic. If

$$d(f^n(x), x) < \alpha$$

for sufficiently small

$$\alpha > 0,$$

then there exists a periodic point

$$p$$

with

$$f^n(p) = p, \qquad d(x, p) < \beta,$$

where

$$\beta \to 0 \quad \textit{as} \quad \alpha \to 0.$$

Proof. Define the periodic pseudo-orbit

$$x_0, x_1, \ldots, x_{n-1}, x_0, \ldots, \qquad x_i = f^i(x).$$

Since

$$d(f^n(x), x) < \alpha,$$

this is an

$$\alpha$$

-pseudo orbit.
By the Shadowing Lemma, there exists a true orbit

$$p_i = f^i(p)$$

such that

$$d(p_i, x_i) < \beta,$$

with

$$\beta \to 0$$

as

$$\alpha \to 0.$$

Because the pseudo-orbit is periodic,

$$x_{i+kn} = x_i \qquad \forall k \in \mathbb{Z}.$$

Hence

$$d(p_{i+kn}, p_i) \leq d(p_{i+kn}, x_{i+kn}) + d(x_i, p_i) < 2\beta.$$

Thus

$$d(f^m(p_{i+n}), f^m(p_i)) < 2\beta \qquad \forall m \in \mathbb{Z}.$$

Choose

$$2\beta < \varepsilon_0,$$

where

$$\varepsilon_0$$

is the expansiveness constant. Then

$$d(f^m(p_{i+n}), f^m(p_i)) < \varepsilon_0 \qquad \forall m,$$

so expansiveness implies

$$p_{i+n} = p_i.$$

Therefore

$$f^n(p) = p,$$

and

$$d(x, p) < \beta.$$

Since

$$\beta \to 0$$

as

$$\alpha \to 0,$$

the result follows. □

Shadowing Lemma for Nonlinear Uniformly Hyperbolic Systems

Theorem 1. *Let*

$$f : \mathbb{R}^m \to \mathbb{R}^m$$

be a C^2 diffeomorphism and let

$$(x_n)_{n\in\mathbb{Z}}$$

be a δ-pseudo orbit:

$$x_{n+1} = f(x_n) + c_n, \qquad \|c_n\| < \delta.$$

Assume that along (x_n) there exists a uniformly hyperbolic splitting

$$\mathbb{R}^m = E_n^s \oplus E_n^u, \qquad Df(x_n)E_n^{s,u} = E_{n+1}^{s,u},$$

and constants $C > 0$, $0 < \lambda < 1$ such that for all $k \geq 0$,

$$\|Df^k(x_n)v^s\| \leq C\lambda^k\|v^s\|, \qquad \|Df^{-k}(x_n)v^u\| \leq C\lambda^k\|v^u\|.$$

Assume the splitting is uniformly transverse. Then for sufficiently small

$$\delta > 0,$$

there exists a unique true orbit

$$(y_n)$$

satisfying

$$y_{n+1} = f(y_n), \qquad \|x_n - y_n\| < \varepsilon$$

for all

$$n \in \mathbb{Z}.$$

Proof. Seek a true orbit of the form

$$y_n = x_n + b_n, \qquad \|b_n\| < \varepsilon.$$

Since

$$y_{n+1} = f(y_n),$$

we obtain

$$x_{n+1} + b_{n+1} = f(x_n + b_n).$$

Using

$$x_{n+1} = f(x_n) + c_n,$$

and Taylor expansion,

$$f(x_n + b_n) = f(x_n) + Df(x_n)b_n + R_n(b_n),$$

where

$$R_n(b_n) = O(\|b_n\|^2),$$

gives

$$b_{n+1} = A_n b_n + R_n(b_n) - c_n,$$

with

$$A_n := Df(x_n).$$

Define

$$g_n(b_n) := R_n(b_n) - c_n.$$

Then

$$b_{n+1} - A_n b_n = g_n.$$

Define the linear operator

$$(Lb)_n := b_{n+1} - A_n b_n.$$

Decompose

$$b_n = b_n^s + b_n^u, \qquad g_n = g_n^s + g_n^u$$

according to

$$E_n^s \oplus E_n^u.$$

The inverse of L is given explicitly by

$$b_n^s = \sum_{k=0}^{\infty} \left(\prod_{i=1}^{k} A_{n-i} \right) g_{n-k-1}^s,$$

$$b_n^u = -\sum_{k=0}^{\infty} \left(\prod_{i=0}^{k} A_{n+i}^{-1} \right) g_{n+k}^u.$$

Uniform hyperbolicity implies

$$\left\| \prod_{i=1}^{k} A_{n-i} v^s \right\| \le C\lambda^k \|v^s\|, \qquad \left\| \prod_{i=0}^{k} A_{n+i}^{-1} v^u \right\| \le C\lambda^k \|v^u\|,$$

so both series converge absolutely and

$$L^{-1}$$

is bounded on bounded sequences.
Define

$$\Phi(b) = L^{-1}(R(b) - c).$$

We show Φ is a contraction on

$$B_\varepsilon = \{b : \|b\|_\infty \le \varepsilon\}.$$

Since

$$R_n(b) = O(\|b\|^2),$$

there exists $K > 0$ such that

$$\|R_n(b)\| \le K\|b\|^2, \qquad \|R_n(b) - R_n(d)\| \le K\varepsilon\|b - d\|.$$

Hence

$$\|\Phi(b)\|_\infty \le \|L^{-1}\|(K\varepsilon^2 + \delta),$$

and

$$\|\Phi(b) - \Phi(d)\|_\infty \le \|L^{-1}\|K\varepsilon\|b - d\|_\infty.$$

Choosing

$$\varepsilon, \delta$$

sufficiently small ensures

$$\Phi(B_\varepsilon) \subseteq B_\varepsilon, \qquad \|L^{-1}\|K\varepsilon < 1,$$

so Φ is a contraction.
By Banach's Fixed Point Theorem, there exists a unique bounded sequence

$$b$$

such that

$$\Phi(b) = b.$$

Equivalently,

$$b_{n+1} = A_n b_n + R_n(b_n) - c_n.$$

Define

$$y_n = x_n + b_n.$$

Then

$$y_{n+1} = x_{n+1} + b_{n+1} = f(x_n + b_n) = f(y_n),$$

so (y_n) is a true orbit. Moreover,

$$\|y_n - x_n\| = \|b_n\| < \varepsilon.$$

Uniqueness follows immediately from Banach's Fixed Point Theorem.

□

Theorem 1 (Shadowing Lemma for Linear Uniformly Hyperbolic Systems)**.** *Let $A : \mathbb{R}^m \to \mathbb{R}^m$ be an invertible linear map admitting a hyperbolic splitting $\mathbb{R}^m = E^s \oplus E^u$, where $A(E^s) = E^s$, $A(E^u) = E^u$, and there exist constants $C > 0$ and $0 < \lambda < 1$ such that for all $n \geq 0$:*

$$\|A^n v^s\| \leq C\lambda^n \|v^s\|, \quad v^s \in E^s \qquad \text{and} \qquad \|A^{-n} v^u\| \leq C\lambda^n \|v^u\|, \quad v^u \in E^u.$$

Then for every $\varepsilon > 0$, there exists $\delta > 0$ such that every δ-pseudo orbit satisfying $\|Ax_n - x_{n+1}\| < \delta$ is uniquely ε-shadowed by a true orbit $y_{n+1} = Ay_n$ such that $\|x_n - y_n\| < \varepsilon$ for all $n \in \mathbb{Z}$.

Proof. **1. Existence via Variation of Constants.** Let $(x_n)_{n\in\mathbb{Z}}$ be a δ-pseudo orbit, written as $x_{n+1} = Ax_n + b_n$ with $\|b_n\| < \delta$. We seek an exact orbit $y_{n+1} = Ay_n$ such that the sequence of deviations $c_n = x_n - y_n$ satisfies the difference equation $c_{n+1} = Ac_n + b_n$. Decomposing all variables along the invariant subspaces $E^s \oplus E^u$, we write:

$$c^s_{n+1} = Ac^s_n + b^s_n \qquad \text{and} \qquad c^u_{n+1} = Ac^u_n + b^u_n, \qquad \text{with } c^{s,u}_n, b^{s,u}_n \in E^{s,u}.$$

We explicitly construct the unique bounded sequence solutions for both components:

$$c^s_n = \sum_{k=0}^{\infty} A^k b^s_{n-1-k} \qquad \text{and} \qquad c^u_n = -\sum_{k=0}^{\infty} A^{-k-1} b^u_{n+k}.$$

We verify the dynamic stepping equations directly:

$$\text{For } E^s: \quad Ac^s_n + b^s_n = \sum_{k=0}^{\infty} A^{k+1} b^s_{n-1-k} + b^s_n = \sum_{k=1}^{\infty} A^k b^s_{n-k} + b^s_n = \sum_{k=0}^{\infty} A^k b^s_{n-k} = c^s_{n+1}.$$

$$\text{For } E^u: \quad Ac^u_n + b^u_n = -\sum_{k=0}^{\infty} A^{-k} b^u_{n+k} + b^u_n = -\sum_{k=1}^{\infty} A^{-k} b^u_{n+k} = -\sum_{j=0}^{\infty} A^{-j-1} b^u_{n+1+j} = c^u_{n+1}.$$

Hence, combining components yields $c_{n+1} = Ac_n + b_n$.

2. Boundedness and ε-Control. Applying the uniform geometric expansion and contraction bounds of the hyperbolic operators alongside the supremum bound on the noise terms ($\|b^{s,u}_n\| \leq \delta$):

$$\|c^s_n\| \leq \sum_{k=0}^{\infty} \|A^k b^s_{n-1-k}\| \leq \sum_{k=0}^{\infty} C\lambda^k \delta = \frac{C\delta}{1-\lambda}.$$

$$\|c^u_n\| \leq \sum_{k=0}^{\infty} \|A^{-k-1} b^u_{n+k}\| \leq \sum_{k=0}^{\infty} C\lambda^{k+1} \delta = \frac{C\lambda\delta}{1-\lambda}.$$

Because the stable and unstable subspaces are uniformly transverse, the projection operators are strictly bounded. Thus, there exists a uniform structural scaling constant $K > 0$ such that:

$$\|c_n\| \leq \|c^s_n\| + \|c^u_n\| \leq \frac{C(1+\lambda)}{1-\lambda}\delta \implies \|c_n\| \leq K\delta.$$

Choosing the noise threshold $\delta < \frac{\varepsilon}{K}$ guarantees the uniform proximity bound:

$$\|x_n - y_n\| = \|c_n\| < \varepsilon \quad \text{for all } n \in \mathbb{Z}.$$

3. Uniqueness. Suppose distinct exact orbits y_n and z_n both ε-shadow the same pseudo-orbit. Their mutual separation sequence $d_n = y_n - z_n$ satisfies the unperturbed linear map $d_{n+1} = Ad_n$. Decomposing via $d_n = d^s_n + d^u_n$, the triangle inequality forces d_n to remain globally bounded across all horizons:

$$\|d_n\| \leq \|y_n - x_n\| + \|x_n - z_n\| < 2\varepsilon \quad \text{for all } n \in \mathbb{Z}.$$

If $d_n^u \neq 0$, forward iteration yields exponential divergence: $\|A^k d_n^u\| \to \infty$ as $k \to \infty$. Alternatively, if $d_n^u = 0$, then $d_n^s \neq 0$, and backward iteration yields exponential divergence: $\|A^{-k} d_n^s\| \to \infty$ as $k \to \infty$. Since unbounded growth contradicts $\|d_n\| < 2\varepsilon$, we must have $d_n = 0$ for all n, meaning $y_n = z_n$. $\square$